ZHAOSEN QU

zqu7@jhu.edu | linkedin.com/in/zmqu | Google Scholar | US Permanent Resident

Research Interest

Ferroelectric and emerging non-volatile memory devices & systems; nanoscale semiconductor fabrication and scaling; thin film synthesis and process integration; emerging 2D & quantum devices

Education

Johns Hopkins University Baltimore, MD
B.S. in Electrical Engineering (GPA: 3.99/4.00) Aug 2024 – Expected: May 2027
Minor in Mathematics
Selected Coursework: Adv. Microfabrication Lab, Optoelectronic Materials & Devices, VLSI, Quantum/Physical Chemistry, Electronic Properties of Materials, Kinetics, Materials Characterization

Research Experience

Dept. of Chemistry, Johns Hopkins University Baltimore, MD
Research Assistant | PI: Prof. Thomas J. Kempa Aug 2026 – Present

- Synthesize bulk and few-layer metal-organic frameworks (MOFs) by chemical vapor deposition (CVD) and transfer them to monolayer TMDs to form heterostructures.
- Study whether chemically tunable MOF lattices imprint periodic potentials onto atomically thin TMDs in 2D MOF-TMD heterostructures.
- Characterize MOF, TMD, and heterostructure morphology and optical response using AFM and optical spectroscopy.

Center for Nanophase Materials Sciences, Oak Ridge National Laboratory Oak Ridge, TN
DOE ORISE Research Student Intern | PI: Dr. Kyle P. Kelley May – Aug 2026

- Optimized oxynitride interlayer on sputtered aluminum nitride (AlN) thin films on SiO_2 and c-axis sapphire (as control) with varying conditions to reduce mosaicity.
- Characterized film mosaicity, crystallinity, and surface roughness primarily through X-ray diffraction (XRD), atomic force microscopy (AFM), and Raman spectroscopy analysis as feedback, showing improved structural qualities with optimized oxynitride treatment.
- This low-temperature (350 °C) sputtering process enables more direct CMOS-compatible photonic integration at scale.

Dept. of Materials Science & Engineering, Johns Hopkins University Baltimore, MD
Research Assistant | PI: Prof. Timothy P. Weihs Dec 2024 – May 2026

- Investigated structure-property relationships in Cu, Cu-Ti, and Ti-V alloys, studying how grain morphology, texture, and defect density govern conductivity and hardness.
- Quantified grain morphology and crystallographic texture using SEM/EBSD and assessed defect structure using XRD, correlating these measurements with Vickers hardness to identify strengthening mechanisms in sputtering-deposited foils.
- Demonstrated composition dependent precipitation strengthening in freestanding Cu-Ti foils, achieving peak hardness of ~350 HV with intermediate Ti contents resulting in the best hardness-conductivity balance.

Dept. of Electrical & Systems Engineering, University of Pennsylvania Philadelphia, PA
NSF Singh REU Intern | PI: Prof. Deep M. Jariwala May – Aug 2025

- Scaled ferroelectric AlScN diode non-volatile memory devices from 50 μm to the 40 nm node, demonstrating viability for high-density (100 $Mbit/mm^2$) storage.
- Characterized ferroelectric switching behavior in sub-50 nm AlScN diodes via P–E hysteresis and I–V measurements, achieving a 20% reduction in coercive field and 10x reduction in leakage current through ITO/Al top contact engineering.

- Investigated 5 nm AlN / 15 nm AlScN heterostructures, showing interfacial coupling mechanisms that enhances polarization, retention, and device reliability via proximity ferroelectricity.

Publications & Manuscripts

4. **Z. Qu***, S. Gellerup*, K. P. Kelley, "Optimization of Nitrided Buffer Layers on Plasma Treated SiO_2 for Reactively Sputtered AlN Films," in preparation for *APL Materials*, 2026.
3. R. Berlia, **Z. Qu**, Y. Sun, R. T. Mehta, and T. P. Weihs, "Combinatorial mapping of composition–processing–property relationships in precipitation-strengthened Cu–Ti alloys," *Materials Science and Engineering: A*, under review, 2026.
2. R. Berlia, C. Singh, **Z. Qu**, R. T. Mehta, and T. P. Weihs, "Revealing bulk-like properties in freestanding PVD copper foils," *Materials Science and Engineering: A*, vol. 947, p. 149163, 2025. doi: 10.1016/j.msea.2025.149163.
1. Z. Hu, H. Cho, R. K. Rai, K. Bao, Y. Zhang, **Z. Qu**, Y. He, Y. Ji, C. Leblanc, K.-H. Kim, Z. Han, Z. Qiu, X. Du, E. A. Stach, R. Olsson, and D. Jariwala, "Demonstration of highly scaled AlScN ferroelectric diode memory with a storage density of 100 Mbit/mm2," *Nano Letters*, vol. 25, no. 37, pp. 13748–13755, 2025. doi: 10.1021/acs.nanolett.5c02961.

Presentations

Z. Qu, S. Gellerup, K. Kelley, "Nitrided Buffer Layers on Plasma Treated SiO2 for Reactively Sputtered AlN," *DOE ORISE Summer 2026 Intern Poster Session*, Oak Ridge, TN, 2026

Z. Qu, Z. Hu, Y. Cao, and D. Jariwala, "Proximity Ferroelectricity in Thin Nitride-Ferroelectric Stack Using 5 nm AlN – 15 nm AlScN," *Yale Undergraduate Research Conference*, New Haven, CT, 2026.

Z. Qu, Z. Hu, Y. Cao, and D. Jariwala, "Overcoming the Memory Wall: Scaling AlScN FeDiodes to sub-50 nm and Enhancing Reliability via Proximity Ferroelectricity," *Singh REU Symposium,* University of Pennsylvania, Philadelphia, PA, 2025

Engineering Experience

Space Radiation Lab, California Institute of Technology — Pasadena, CA
Electrical Engineering Intern | Mentor: Jill Burnham — Jun – Jul 2023

- Diagnosed and repaired overstressed components in the SPHEREx flight assembly by performing root-cause analysis and schematic review in Altium Designer, restoring system reliability for NASA mission qualification.
- Conducted accelerated stress-testing on circuit components using programmable power supplies and oscilloscopes to characterize failure modes.
- Managed Bill of Materials (BOM) revisions across multiple flight board versions, ensuring traceability and configuration control for mission-critical hardware.

Teaching Experience

Johns Hopkins University — Baltimore, MD

- *Head Course Assistant: Intermediate Programming (C/C++)* — Aug 2026 – Present
- *Course Assistant: Intermediate Programming (C/C++)* — Aug 2025 – May 2026
- *Teaching Assistant: Linear Algebra* — Aug 2025 – May 2026
- *Course/Learning Assistant: Calculus II, Physics II* — Jan 2025 – May 2025

TECHNICAL SKILLS

Synthesis & Fabrication: Reactive Magnetron Sputtering, Chemical Vapor Deposition (CVD), Reactive Ion Etching
Characterization: X-ray diffraction, SEM, EBSD, Raman Spectroscopy, Atomic Force Microscopy (AFM), Electrical Characterization (Keithley SMUs; I-V; P-E)
Programming & Software: Python, C, C++, Java